

MO7019: Statistical Methods for Climate Science

Statistika Metoder For Klimatvetenskap (7.5 hp)

Exercises

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Chap. 1

Exercise 1

Coin tossing (Bernoulli).

Let X be a random variable with two outcomes $X = \{H, T\} = \{0, 1\}$, with probabilities $Pr(X = 0) = p$ and $Pr(X = 1) = q$.

Calculate the mean and variance of X .

Exercise 2

Same as Exercise 1 but for the Poisson distribution.

Exercise 3

The Gaussian random variable has the following probability density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

Calculate the mean and variance of X .

Calculate the skewness (3d order moment) and kurtosis (4th order moment) of X .

Extra exercise

Bivariate normal

Let (X, Y) be a bivariate normal distribution, and let $\rho = \text{corr}(X, Y)$ (correlation coefficient between X and Y), $\sigma_x^2 = \text{var}(X)$, and $\sigma_y^2 = \text{var}(Y)$. Denote by Σ the covariance matrix of (X, Y) .

Calculate the joint distribution $f(x, y)$ of (X, Y) .

What is the nature of the marginal distributions $f_X(x)$ and $f_Y(y)$ of X and Y respectively.

Derive the conditional probability distribution of $Y|X = x_0$.

Conclude.

Chap. 2

Exercise 1

Let $x_1, x_2, \dots, x_n$ be a real time series with zero mean and unit variance (centered). We model it with a AR(1) process:

$$x_{t+1} = \alpha x_t + \varepsilon_{t+1}$$

Derive the value of α and the variance σ^2 of ε_t .

Exercise 2

Consider the same time series of Exercise 3, show that the coefficients α_1 and α_2 of a AR(2) model satisfy:

$$x_{t+2} = \alpha_1 x_{t+1} + \alpha_2 x_t + \varepsilon_{t+2}.$$

satisfy:

$$\begin{pmatrix} 1 & \rho(1) \\ \rho(1) & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \end{pmatrix}$$

where $\rho(\tau)$ is the autocorrelation function of x_t , at lag τ .

Can you anticipate the equation to determine the coefficients of a AR(3) model and in general of a AR(m) model:

$$x_{t+m} = \alpha_1 x_{t+m-1} + \dots + \alpha_m x_t + \varepsilon_{t+m}.$$

Chap. 3

Exercise

We consider a sample of two time series x_t and y_t , $t = 1, \dots, 6$

	1	2	3	4	5	6	Mean	Standard deviation
x	1.15	-7.7	1.41	-0.74	-0.24	-1.25	-1.2	1.1
y	0.91	0.59	0.35	1.25	0.92	-0.23	0.63	0.52

Table 1. Example of time series x_t and y_t , $t = 1, \dots, 6$.

We want to check whether the (unknown) means μ_x and μ_y of x_t and y_t are different from zero.

Assume $\alpha = 5\%$, apply the appropriate statistical test to test the null hypothesis H_0 :

$H_0 : \mu_x = 0$ versus $H_1 : \mu_x \neq 0$ for x_t , and

$H_0 : \mu_y = 0$ versus $H_1 : \mu_y \neq 0$ for y_t .

Next we want to test the hypothesis :

$H_0 : \mu_x = \mu_y$ versus $H_1 : \mu_x \neq \mu_y$. Using again $\alpha = 5\%$ discuss whether the null hypothesis can be rejected.

Comment on the result.

Chap. 5

Exercise 1

Consider the regression equation:

$$y_i = \alpha + \beta x_i + \varepsilon_i, \quad i = 1, \dots, n$$

where the sample y_t , $t = 1, \dots, n$ is drawn from a normal distribution, and assume that $\bar{x} = \bar{y} = 0$. We let $\hat{\beta}$ the estimator of β based the available sample.

1. Calculate $E(\hat{\beta})$
2. show that $\text{var}(\hat{\beta}) = \frac{\sigma^2}{\sum x_i^2} = \frac{\sigma^2}{S_{xx}}$ where $\sigma^2 = E(y_i^2)$.
3. Conclude on the distribution of $\hat{\beta}$.

Exercise 2

Show that the coefficient of determination R^2 is the square of the correlation coefficient between x_t and y_t , $t = 1, \dots, n$:

$$R^2 = \frac{SSR}{SST} = r_{xy}^2$$

Exercise 3: Application to NAO, PNA and Nino-3

Consider the following time series of the North Atlantic oscillation (NAO), Pacific North America (PNA), and Nino-3 indices for January over the 12 years 2012-2023 (source: Physical Science Lab, NOAA, psl.noaa.gov/data/climateindices/list).

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
NAO	2.05	1.08	-0.17	1.57	-0.37	0.05	1.17	0.18	1.05	-1.8	0.74	0.95
PNA	0.13	0.05	0.59	0.14	1.94	-0.29	-0.14	0.42	-0.95	-0.41	0.65	-0.37
Nino-3	-0.57	-0.66	-0.43	0.35	2.59	-0.08	-1.18	0.6	0.22	-0.62	-1.31	-0.55

Table 2. January mean time series f NAO, PNA and Nino-3.

1. Fit a regression model linking PNA to Nino-3, i.e. $PNA = aNino - 3 + \varepsilon$. Find the p-value of the regression coefficient a . Is it significant at 10% level?
2. Fit a regression model linking NAO to PNA and Nino-3, i.e. $NAO = bPNA + cNino - 3 + \varepsilon$. Are b and c significant at 10% level?
3. Can you anticipate a potential problem with the model of question 2? Can you think of a solution?

Chap. 6

Exercise 1

Let $\mathbf{X}$ be a data matrix,

$$\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)^T = \begin{pmatrix} x_{1,1} & \dots & x_{1,p} \\ \vdots & & \vdots \\ x_{n,1} & \dots & x_{n,p} \end{pmatrix}$$

where n is the sample size and p the number of variables (e.g. grid points). We assume that the matrix is centered, i.e. $\sum_{t=1}^n x_{t,k} = 0$. Show that the sample covariance is $\mathbf{S} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X}$

Exercise 2

Let $\mathbf{a}$ be p -dimensional vector.

Show that the variance of the data projected onto $\mathbf{a}$ is $v = \mathbf{a}^T \mathbf{S} \mathbf{a}$.

Find the direction that maximises the variance v .

Exercise 3

Which of the following are valid covariance matrices, and why?

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 1 & -1 \end{pmatrix},$$

Chapters 12 & 13: ML and neural networks

Exercise 1

Fisher LDA. Consider two classes with respective means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ and covariances $\mathbf{S}_1$ and $\mathbf{S}_2$. The between group variance along a direction $\mathbf{a}$ is $(\mathbf{a}^T(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1))^2$, and the within-group variance is $\mathbf{a}^T(\mathbf{S}_1 + \mathbf{S}_2)\mathbf{a}$.

Find the separating hyperplane between the two groups.

Exercise 2

K-means and hierarchical clustering. Explain the difference between k-means and hierarchical clustering, and highlight the effect of the various types of dissimilarity measures in the latter.

Exercise 3

Perceptron. Consider a perceptron with nonlinear sigmoid activation function:

$$o = \sigma(\mathbf{a}^T \mathbf{x})$$

Can this perceptron do binary classification of nonlinearly separable classes?

If not what can be done?

Exercise 4

NN with one hidden layer. Consider a neural network with one hidden layer with two units with sigmoid function and one single linear output unit. Assume a 2-dimensional input space. Calculate the output y and the gradient of the costfunction, assuming n training input patterns $\mathbf{x}_i$, and targets t_i , $i = 1, \dots, n$.

Write down the updating rule of the neural net weights.